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The application of virtual reality in food consumer behavior research: A systematic review

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ABSTRACT

Background: The development of virtual reality (VR) technology has the potential to provide enormous opportunities for food and consumer behavior research to further explore current research methodologies and establish new ones. VR is different from the environment we experience in daily life. Therefore, the extent to which VR can be applied in food consumer behavior research needs to be assessed on the basis of the evidence provided by the studies that have employed VR.

Approach: To better understand VR technology and its potential applications and limitations in food and consumer behavior research, we conducted a systematic review of the literature on VR in food and consumer behavior research. The applications of VR addressed in the articles were extracted. The validity of VR for food and consumer behavior research was analyzed.

Key findings: VR has been used to build complex and realistic contextual environments for conducting food sensory evaluation research. VR was also used as a substitute (e.g., a VR buffet and supermarket) to real life settings for investigating consumers' purchasing behavior and food choices. Virtual food and food-related cues were used to induce psychological and physiological responses in consumers. The results supported the validity of VR as a tool for investigating consumers' behavior toward food.

1. Introduction

Virtual reality (VR) is defined as "a computer-generated digital environment that can be interacted with as if that environment were real" (Jerald, 2015, p. 9). The applications of VR technology have drawn the attention not only of the entertainment industry (e.g., video games, theme parks, immersive film) but also of scientific researchers. VR has been used in many research fields, such as data visualization, architecture, aviation simulation, and product prototyping (Jerald, 2015). Because of its potential to create realistic and engaging environments and experiences, it has been employed to provide safe environments for studies that would be dangerous or unethical in real life (RL). For example, it has been applied in confrontation therapy to treat anxiety or phobias (e.g., acrophobia or arachnophobia) by exposing people to feared stimuli without actually placing them in the feared environments (Choi, Jang, Ku, Shin, & Kim, 2001; Muhlberger, 2003).

VR has also been applied in the research fields of food and consumer behavior. The advantage of using VR to observe consumer behavior is that it can provide a high level of standardization that is similar to the level achieved in laboratory experiments (Hartmann & Siegrist, 2019). At the same time, experimental environments with highly realistic contextual information for point-of-sale and point-of-consumption can be conveniently designed or built in VR. Subsequently, the limitations of traditional test environments (e.g., field and laboratory), such as environmental cues that cannot be fully controlled and low ecological validity can be overcome (Xu, Demir-Kaymaz, Hartmann, Menozzi, & Siegrist, 2021). It allows researchers to investigate consumers' behavior in an environment that is as realistic as an RL situation in which the behavior naturally occurs while environmental cues can simultaneously be fully controlled. VR seems to be promising and beneficial for establishing alternative research methodologies in food and consumer behavior research, especially for research questions that would be arduous to investigate in RL—for example, the measurement of the impact of environmental factors in a physical supermarket on consumers' purchasing behavior and food choices.

Even though VR may provide manifold opportunities, it is still a computer-generated artificial environment that is different from the environments we experience in daily life. Consequently, to what extent

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VR can be employed in food and consumer behavior research and whether it is a valid tool for collecting consumers' behavior data need to be examined. Although some studies have provided evidence to support the validity of VR as a data collection tool for consumer behavior research (Pizzi, Scarpi, Pichierri, & Vannucci, 2019; Siegrist et al., 2019; Ung, Menozzi, Hartmann, & Siegrist, 2018; van Herpen, van den Broek, van Trijp, & Yu, 2016; Waterlander, Jiang, Steenhuis, & Ni Mhurchu, 2015; Xu et al., 2021), comprehensive analyses based on systematically reviewed data are needed. Thus far, no systematic review papers have focused on answering these questions. Therefore, the aim of the present review was to systematically inspect the types of food and consumer research that VR technology has been employed in and to provide evidence-based information about the validity and the limitations, but also the possibilities, of VR.

This systematic review pursues the following research objectives: 1) to determine which research domains within food and consumer behavior research VR has been employed in, and what type of VR systems were typically used in each research domain; 2) to investigate whether VR is a valid tool for food and consumer behavior research based on the comparison between VR settings and RL settings; and 3) to identify the gaps that exist in the literature and to inform future experimental and applied practice in the field of food and consumer behavior. The results of this systematic review paper are important for researchers or food industries who aim to gain more knowledge of the technology and use it in their future studies or in, for example, product design and marketing research.

2. Theoretical background

2.1. Measuring the validity of VR

Two approaches to measuring the validity or usability of VR were found in the literature. The first is the use of post-VR-experience questionnaires to measure consumers' subjective ratings of the usability or validity of the VR system (Allman-Farinelli et al., 2019; Isgin-Atici et al., 2020). The most frequently employed questionnaire in the literature is the system usability questionnaire originally developed by Brooke (1996). A usability score was usually calculated based on the participants' responses. The higher the usability score, the higher the usability or validity of the VR application. The other approach, which is most frequently used, is to compare consumer behavior data collected in VR settings and in RL settings (Ammann, Stucki, & Siegrist, 2020; Cheah et al., 2019, 2020; Lombart et al., 2020; Persky, Goldring, Turner, Cohen, & Kistler, 2018; Pizzi et al., 2019; Siegrist et al., 2019; Ung et al., 2018; van Herpen et al., 2016; Waterlander et al., 2015; Xu et al., 2021). Under this approach, the validity of VR can be supported if the consumer behavior data in VR are highly comparable and correlate with the data collected in RL or other well-recognized experimental settings. In contrast to the first approach, this approach allows consumers' behavior data collected in different environments or settings to be objectively observed and compared, and objective evidence therefore can be expected.

In this systematic review, the studies that aimed at investigating the comparability or similarity of consumers' behavior in VR and RL settings were examined and summarized into a table (Table 2) to fulfill the research objective of determining the validity of VR. Namely, studies that employed both VR and RL settings to collect data but without the goal of exploring their comparability were not included. For example, some sensory evaluation studies have employed VR to provide contextual information and compared the results with those obtained from a traditional sensory booth; however, their goals were to study the differences created by contextual information rather than whether the two settings were comparable (Andersen, Kraus, Ritz, & Bredie, 2019; Bangcuyo et al., 2015; Hannum, Forzley, Popper, & Simons, 2019; Hathaway & Simons, 2017; Kong et al., 2020; Sinesio et al., 2019; Torrico et al., 2020, 2021; Worch et al., 2020). Studies that have compared other data, such as consumers' emotional responses and salivation, instead of behavioral data collected in VR and RL settings were also not included (Gorini, Griez, Petrova, & Riva, 2010; Gutierrez-Maldonado, Pla-Sanjuanelo, & Ferrer-Garcia, 2016; Ledoux, Nguyen, Bakos-Block, & Bordnick, 2013; van der Waal, Janssen, Antheunis, Culleton, & van der Laan, 2021).

2.2. VR systems and immersion

Schoor and colleagues have classified VR systems into the categories of fully immersive, semi-immersive, and low-immersive (desktop VR) systems (Schoor et al., 2009). As indicated by their names, different VR systems can provide different immersion levels. Immersion is a description of the extent to which the VR system can deliver "an inclusive, extensive, surrounding and vivid illusion of reality to the senses of a human participants" (Slater & Wilbur, 1997, p. 2). It is an objective factor that indicates the potential of the VR system or application itself to engage participants. However, how the participants subjectively perceived the immersion is known as presence, which means a sense of "being" in the virtual environment (Jerald, 2015). Immersion and presence together describe how engaging or realistic a VR experience is objectively and subjectively.

The level of immersion a VR system or application can provide depends on its design. According to the definition of immersion, the design of the VR system must be surrounding, inclusive, vivid, and extensive to provide a high level of immersion. High surroundness can be achieved by a wide field of view, which depends on the display of the VR system. For example, the highest field-of-view can be provided in a fully immersive VR system (e.g., a head-mounted-display [HMD] system or a cave automatic virtual environment [CAVE] system), because participants are blocked from the outside world and only experience and interact with the displayed virtual environment. In a semi-immersive VR system (e.g., video wall system), on the other hand, participants may not be fully blocked from the real world, and most of the field-of-view is provided by hardware such as video projectors. As for the low-immersive system (e.g., desktop VR), the virtual environment is normally displayed on a computer screen. Therefore, only a limited field-of-view can be provided (Narciso et al., 2021). The inclusiveness level can be affected by interactability, which is the user's capability to interact with and influence the virtual environments and events, and the locomotion, which is the way of moving around in the virtual environments, in the VR systems (Jerald, 2015). Vividness can be influenced by the quality of the virtual content simulated: the more realistic the content, the higher the vividness. Extensiveness refers to the range of sensory modalities (e.g., visuals, audio, and haptic feedback) presented to the participants: the wider the range of sensory modalities provided, the higher the extensiveness level. Fig. 1 shows a detailed graphic illustration of the design of VR systems that can provide different immersion levels.

For ease of comprehensibility, the VR systems mentioned in this systematic review are represented by the displays they use. For example, a VR system with a head-mounted display (HMD), controllers to provide interaction, and physical walking for locomotion is called an HMD system. A VR system with a computer screen/desktop as a display, a keyboard and mouse for interaction, and no locomotion is called a desktop VR system. The immersion level in this systematic review refers to the objective immersion (not presence).

3. Methods

3.1. Literature search

A literature search was conducted on the Web of Science (databases included Core Collection, Current Contents Connect, FSTA—the food science resource, and MEDLINE) and Scopus in April 2021. The following search string was used: (("virtual reality" OR "virtual

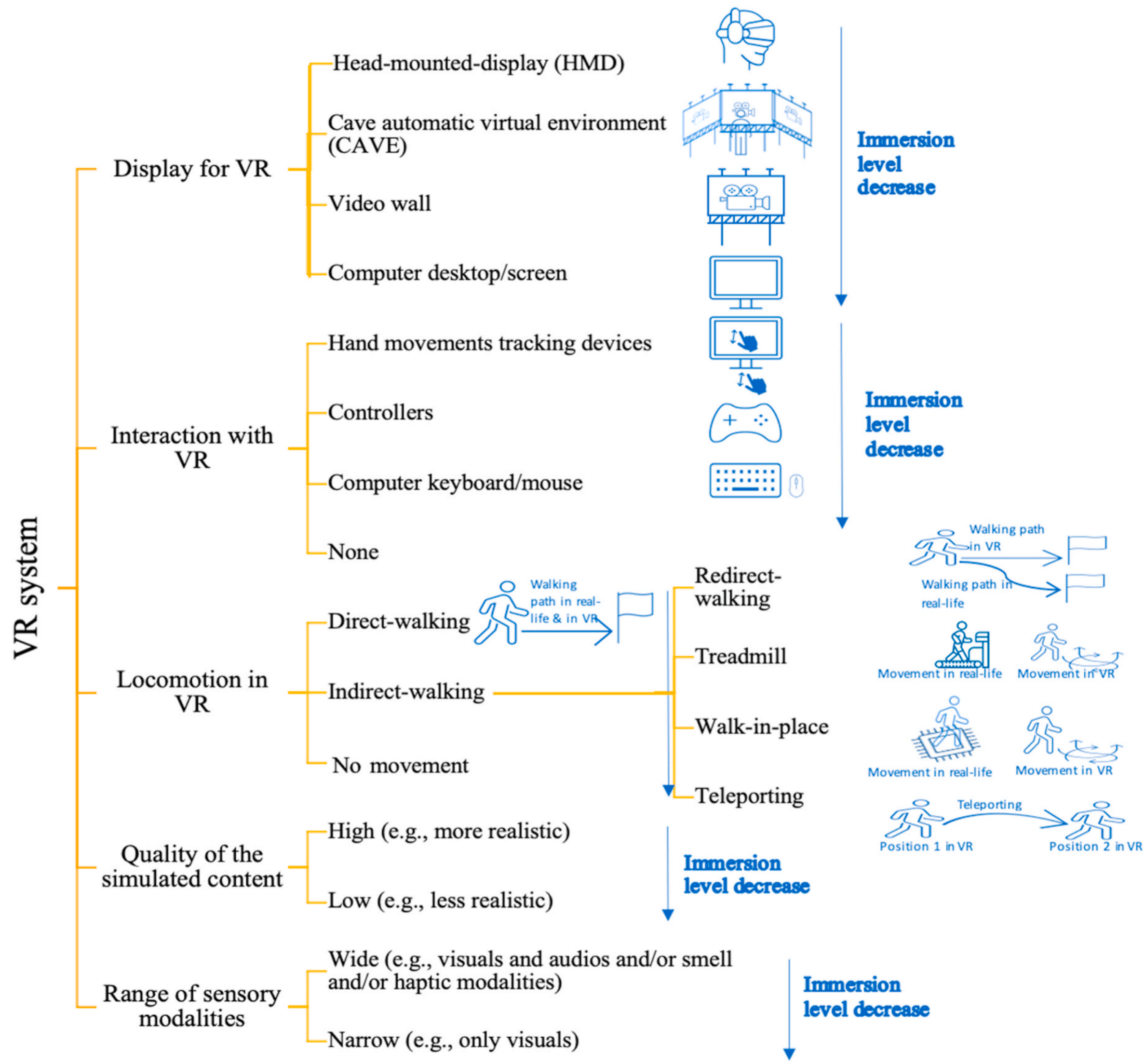

Fig. 1. The designs of virtual reality (VR) systems that can provide different immersion levels.

environment" OR "immersive environment") AND ((food*) OR (consum*) OR (eat*))). Only peer-reviewed journal articles and conference proceedings were included. The search was restricted to English language research articles from 2005 to 2021. This date criterion was chosen because VR during the 1980s and 1990s was still in the exploration phase, and its applications were mainly focused on professional research (e.g., aviation simulation, military use) and entertainment. In addition, VR-related research was not widely published in the first few years of the 21st century (Jerald, 2015).

3.2. Selection and screening

The literature search resulted in 9230 records. After the removal of duplicate records, the records were screened by title and abstract first. Two screeners performed this procedure independently. The inter-rater reliability between the two screeners was checked, which yield an average agreement of 91%. The disagreements between the screeners were solved by discussion. The full-text publication was reviewed if the articles met the inclusion criteria in the title and abstract screening process. Overall, 6970 articles were excluded because they did not meet the inclusion criteria described in Table 1. Specifically, these articles were not related to food and consumer behavior, did not use VR (e.g., investigating video games instead of VR), or did not report primary quantitative results (e.g., review articles or qualitative studies). This exclusion resulted in 137 articles whose full text was screened. Using the same criteria described for the screening of the titles and abstracts, 52 articles were excluded because they did not meet the inclusion criteria. This resulted in a set of 85 articles that were included in the systematic

Table 1
The inclusion and exclusion criteria used for article selection.

Inclusion	Exclusion
Studies that used VR (i.e., fully immersive, semi-immersive, and low-immersive VR systems) in research related to food and consumer behavior toward food	Studies that used other 3D (immersive) systems or environments such as augmented reality (AR) and video games
	Studies related to other research topics within consumer behavior, such as marketing, tourism, and hotel management
	Studies that investigated the therapeutic, pedagogical, and psychological applications of VR
	Literature review, qualitative studies (e.g., in-depth interviews), abstracts, prototypes or concepts, and gray literature

review. A PRISMA flow diagram compiling the study selection process is shown in Fig. 2 (Page et al., 2021).

4. Results

4.1. Characteristics of the included studies

There are 85 articles included in this systematic review, with the date of publication ranging from 2005 to April 2021. The number of included articles is higher in later years, and most of the articles (91%) were published from 2015 to April 2021. The summary can be found in Fig. 3. Relatively young samples were tested in the studies. The mean age of the samples in most of the articles (n = 45) was younger than 45 years old. There was no article tested with samples older than 65 years old (mean age), and only one study tested children (Karkar, Salahuddin, Almaadeed, Aljaam, & Halabi, 2018). The gender distributions of the samples were also examined. In general, samples in most of the articles (n = 63) included 50% or more female participants. There were 12 articles that only tested female participants. A few articles (n = 9) had samples with more male participants. Among these, two articles only used male participants (Alkahtani, Eisa, Kannas, & Shamlan, 2019; Worch et al., 2020). An examination of the samples' place of residence showed that most of the studies (n = 41) recruited samples in European countries. Among them, samples from Spain and Italy were most frequently used, followed by Switzerland and the Netherlands. Studies (n = 25) also often recruited samples from the USA. In addition, a few

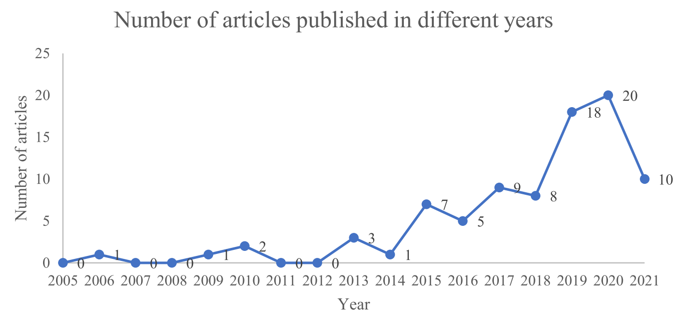

Fig. 3. The number of included articles published in different years from 2005 to April 2021. The literature search was limited to articles that were published from 2005 to April 2021.

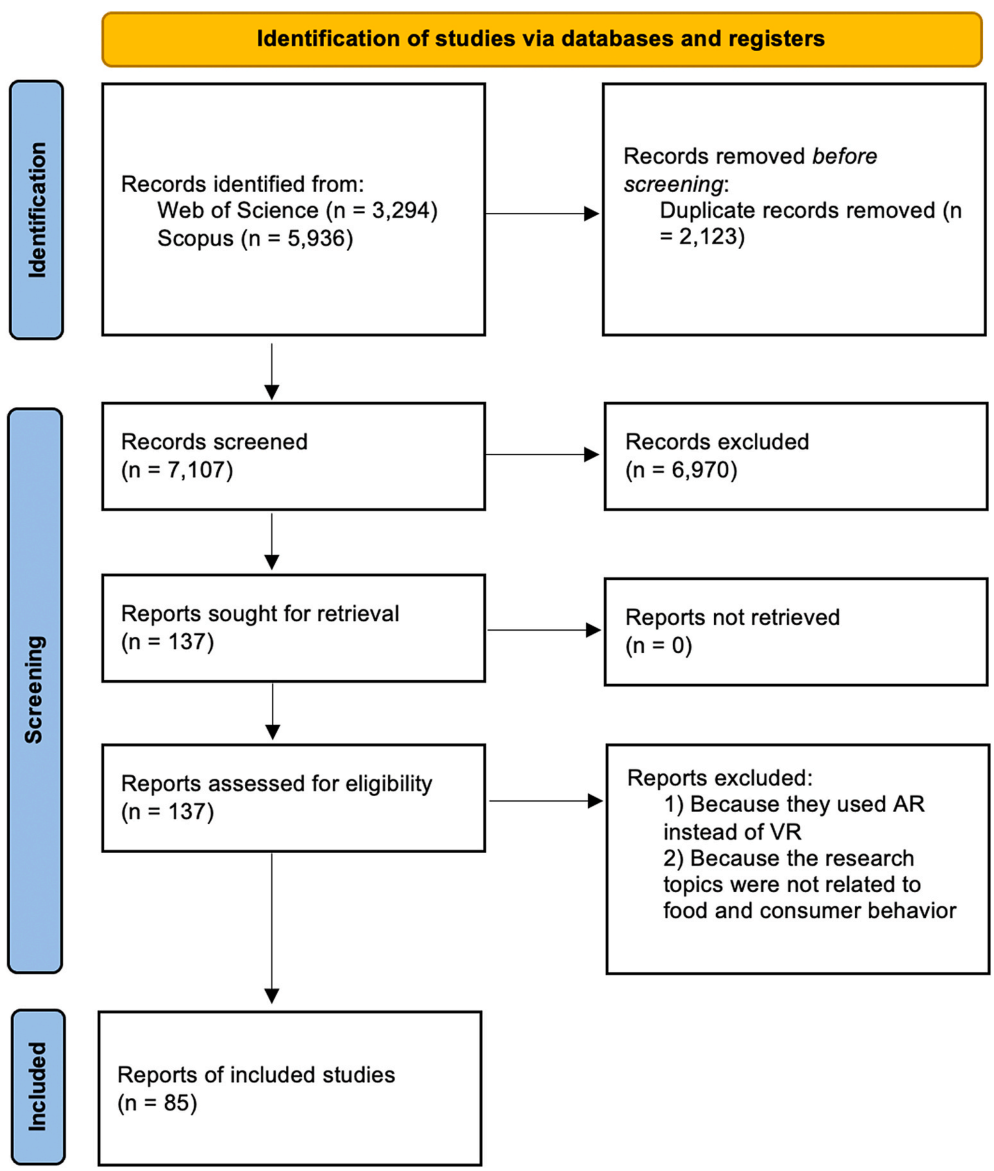

Fig. 2. PRISMA flowchart of the study selection processes of the systematic review.

studies also recruited samples from New Zealand, China, Australia, and Japan. For detailed information about all the reviewed articles, please refer to Table 1 in the supplementary material.

4.2. Research domain and test stimuli (food)

Each of the articles included was examined for its topic and research domain based on its research aims, questions, and interventions. Three main research domains were identified: 1) sensory evaluation of food products in VR; 2) exposures and responses to virtual food or food-related cues and environments; and 3) consumer behavior towards food in VR settings. The third research domain can be further classified into the following categories: 1) shopping behavior and product perception in VR supermarket settings; 2) food choice in VR buffet settings; and 3) other applications. Fig. 4 shows the percentage of each research domain among all included articles.

4.2.1. The application of VR in studying food sensory evaluation

This research domain accounted for 24% of the included articles. Studies in this research domain examined the applications of VR on providing contextual information or environments. They compared participants' sensory evaluation of the same food stimuli in test environments with or without contextual information (e.g., VR contextual environment, pictures contextual environment, RL contextual environment, and traditional sensory booth) (Bangcuyo et al., 2015; Hannum et al., 2019; Hathaway & Simons, 2017; Kong et al., 2020; Oliver & Hollis, 2021; Pennanen, Närväinen, Vanhatalo, Raisamo, & Sozer, 2020; Picket & Dando, 2019; Sinesio et al., 2019; Stelick, Penano, Riak, & Dando, 2018; (Torrico et al., 2020; Torrico et al., 2021)). Bangcuyo et al. (2015) indicated that participants felt significantly more engaged and immersed in virtual environments with contextual information than in traditional sensory booths. Studies also found that different contexts elicited different emotions associated with the evaluated food products (Kong et al., 2020; Torrico et al., 2020, 2021). The results of some studies also acknowledged that contexts did influence the evaluation of some products, such as wine, coffee, and snacks (Bangcuyo et al., 2015; Barbosa Escobar, Petit, & Velasco, 2021; Chen, Huang, Faber, Makransky, & Perez-Cueto, 2020; Hathaway & Simons, 2017; Nivedhan, Mielby, & Wang, 2020; Torrico et al., 2020). However, the magnitude of the impact of VR depends on the products that are examined and the properties that are evaluated. For example, changing the evaluation environment can alter the perception but not necessarily the acceptability of a wine sample (Torrico et al., 2020).

The other studies investigated the impact of manipulating the sensory characteristics of food on evaluation and perception of food. Three studies investigated the impact of VR-based color simulation of drinks (i.e., tea, coffee, and juices) on sensory evaluation of the drinks (Ammann, Stucki, & Siegrist, 2020; Huang, Huang, & Wan, 2019; Wang, Meyer, Waters, & Zendle, 2020). Ammann, Stucki, and Siegrist (2020) showed that participants had more difficulty identifying the flavor of a product when it was presented in a color that have modified from the original. Wang et al. (2020) suggested that viewing a simulated coffee color in VR did influence perceived creaminess but not perceived sweetness or liking. Huang et al. (2019) found that the simulation of the actual color of teas did not influence the taste ratings, but the color that participants associated with the taste of the teas influenced their saltiness rating of the teas. A study conducted by Huang and colleagues tested the effects of packaging color-flavor incongruency of potato chips on the experience of the products and perception of the brands (Huang & Wan, 2019). They also explored the individual differences in the reactions to the color–flavor incongruency (Huang, Zhao, & Wan, 2021). The researchers found that participants tend to like a food product less if the color of its packaging is incongruent with its flavor label, although the brand in such cases was perceived as more innovative. Sugita, Zempo, Mizutani, and Wakatsuki (2017) examined whether a chromatic gradient color transition of a self-made fizzing candy through an HMD would modify impressions and evaluations of it. They found that the impression can be altered by changing the visual information (color) delivered to the participants. These studies showed that manipulating characteristics of food products in VR can influence perception and liking of the products.

4.2.2. The application of VR in studying food (-related) cue exposures and responses

The studies in this research domain accounted for 28% of all included articles. They mainly investigated the impact of exposure to

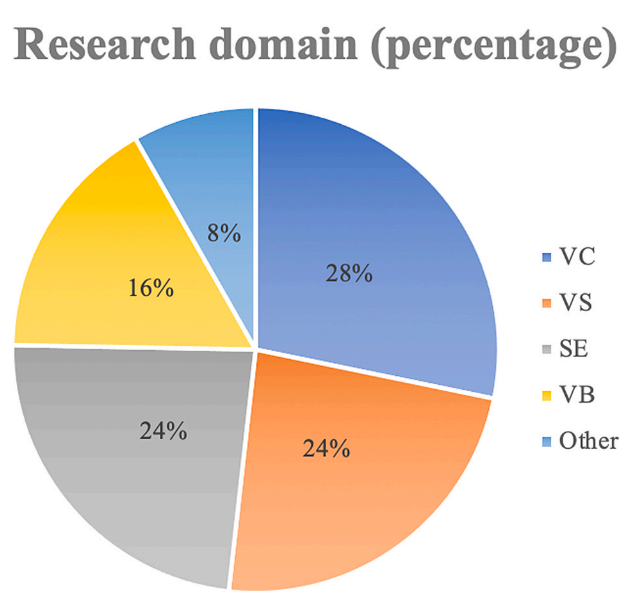

Fig. 4. Research domains in which virtual reality has been employed and the percentage of each research domain within all reviewed articles. VC: exposures and responses to virtual food or food-related cues and environments; VS: shopping behavior and product perception in VR supermarket settings; SE: sensory evaluation of food products in VR; VB: food choice in VR buffet settings.

virtual food or food-related cues and environments on participants' psychological (e.g., food craving, anxiety level, depression episode) or physiological responses (e.g., salivation). Virtual food or food-related cues and environments have been used as alternatives to RL cues in cue exposure treatment or inhibitory control training for treating eating disorders (Ferrer-Garcia et al., 2014, 2017, 2013; Ferrer-García, Gutiérrez-Maldonado et al., 2017; Ferrer-García, Gutiérrez-Maldonado, Caqueo-Urízar, & Moreno, 2009; Ferrer-Garcia, Gutierrez-Maldonado, Treasure, & Vilalta-Abella, 2015; Gorini et al., 2010; Gutiérrez-Maldonado, Ferrer-García, Caqueo-Urízar, & Moreno, 2010; Gutierrez-Maldonado et al., 2016; Gutiérrez-Maldonado, Ferrer-García, Caqueo-Urízar, & Letosa-Porta, 2006; Manasse, Lampe, Juarascio, Zhu, & Forman, 2021; Perpiñá et al., 2013; Pla-Sanjuanelo et al., 2017, 2019, 2015; Tuanquin, Hoermann, Petersen, & Lindeman, 2018). Participants' food craving level, anxiety level, and depressed mood episodes were used as outcome measures to test the effectiveness of virtual cue exposures. These studies either compared the effectiveness of different virtual food cues (e.g., high-calorie virtual food and low-calorie virtual food) or compared the effectiveness of virtual food cues with real food cues and picture food cues. They found that virtual food or food-related cues and environments can successfully induce food craving, anxiety, and depressed mood (Ferrer-Garcia et al., 2014, 2017, 2013; (Ferrer-García, Gutiérrez-Maldonado et al., 2017, Ferrer-García et al., 2009); Ferrer-Garcia, Gutierrez-Maldonado, Pla-Sanjuanelo et al., 2015; Ferrer-Garcia, Gutierrez-Maldonado, Treasure, & Vilalta-Abella, 2015; Gutierrez-Maldonado et al., 2016; Gutiérrez-Maldonado et al., 2010, 2006; Perpiñá et al., 2013; Pla-Sanjuanelo et al., 2017, 2019, 2015; Tuanquin et al., 2018). Also, Gorini et al. (2010) found that virtual food cues could elicit similar anxiety levels in participants as those elicited by real food cues and higher anxiety levels than those elicited by picture cues. Manasse et al. (2021) suggested that participants had significnatly fewer loss-of-control episodes after exposure to the VR cues in inhibitory control training.

Some studies also aimed at testing participants' psychological or physiological responses to virtual food or food-related cues and environments. However, they focused on the general population instead of subjects with eating disorders (Alkahtani et al., 2019; Ammann, Hartmann, Peterhans, Ropelato, & Siegrist, 2020; Andersen et al., 2019; Kuo, Lee, & Chiou, 2016; Ledoux et al., 2013; van der Waal et al., 2021; Worch et al., 2020). Among these articles, four compared the participants' responses to virtual, picture, and real food or food–related cues or environments (Andersen et al., 2019; Ledoux et al., 2013; van der Waal et al., 2021; Worch et al., 2020). Two other studies investigated the influence of exposure to virtual cues on participants' eating behavior afterward (Alkahtani et al., 2019; Kuo et al., 2016). Ammann and colleagues tested the ability of VR food-related cues to induce disgust (Ammann, Hartmann, et al., 2020). These studies showed that virtual cues can successfully induce psychological responses (e.g., food craving and disgust) (Alkahtani et al., 2019; Ammann, Hartmann, et al., 2020; Andersen et al., 2019; Kuo et al., 2016; Ledoux et al., 2013; van der Waal et al., 2021; Worch et al., 2020). The results also showed that virtual food cues, compared to non-food cues, are more comparable to real-food cues in eliciting food craving. However, the influence of exposure to virtual food cues on physiological responses (e.g., salivation) was not significant (van der Waal et al., 2021).

4.2.3. The application of VR supermarket in studying consumers' shopping behavior

Of the reviewed articles, 24% investigated consumers' shopping behavior and product perception in VR supermarket settings (Bigné, Llinares, & Torrecilla, 2016; Blitstein, Guthrie, & Rains, 2020; Blom, Gillebaart, De Boer, van der Laan, & De Ridder, 2021; Goedegebure, van Herpen, & van Trijp, 2020; Hoenink et al., 2020; Liu, Hooker, Parasidis, & Simons, 2017; Lombart et al., 2019, 2020; Payne Riches, Aveyard, Piernas, Rayner, & Jebb, 2019; Peukert, Pfeiffer, Meißner, Pfeiffer, & Weinhardt, 2019; Pizzi et al., 2019; Rodriguez-Raecke, Sommer, Brünner, Müschenich, & Sijben, 2020; Schnack et al., 2019, Schnack, Wright, & Holdershaw, 2020; Siegrist et al., 2019; van Herpen et al., 2016; (Verhulst, Normand, Lombart, & Moreau, 2017; Verhulst, Normand, Lombart, Sugimoto, & Moreau, 2018; Waterlander et al., 2015; Zhao, Huang, Spence, & Wan, 2017). Specifically, some studies compared participants' food selection or purchasing behavior in an immersive virtual simulated store with participants' food selection and purchasing behavior in a physical store (Pizzi et al., 2019; Siegrist et al., 2019; van Herpen et al., 2016; Waterlander et al., 2015) or in a desktop supermarket, a virtual simulated store with a lower immersion level (Peukert et al., 2019; Schnack et al., 2019). Their results showed that participants' food selection and shopping behavior in the VR stores was comparable to those in the RL stores. Studies have examined the effect of the shape of food products, packaging, and labels on the food packages or shelf-tags (i.e., a popularity label, "all-natural" label, or front-of-package nutrition information label) on consumers' perceptions and purchasing behavior towards the tested food items (Blitstein et al., 2020; Goedegebure et al., 2020; Liu et al., 2017; Lombart et al., 2019; (Lombart et al, 2020; Verhulst et al., 2017); Zhao et al., 2017). The impact of embodying participants of a normal body-mass-index in an obese or a normal virtual body on their perception and purchase behavior of food products was also investigated in one study (Verhulst et al., 2018). Bigné et al. (2016) combined VR with eye-tracking and human behavior tracking technologies to study the effect of store navigation and gaze behavior on consumers' purchase decisions. The other two studies investigated the efficacy of nudging strategies (e.g., making healthier food alternatives more salient) on promoting healthy food purchases (Blom et al., 2021; Hoenink et al., 2020). Blom et al. (2021) found that the healthy food purchases increased in the nudging condition (vs. control condition). Hoenink et al. (2020) investigated the efficacy of combining nudging and pricing strategies on increasing healthy food purchases. The researchers found that nudging and non-salient pricing alone did not statistically significantly increase healthy food purchases; however, combining salient pricing and nudging increased healthy food purchases.

4.2.4. The application of VR buffets in studying food choice

Of the included articles, 16% belong to this research domain. Some studies compared participants' food choices in a VR buffet to participants' food choices in an RL buffet or a fake food buffet, and found the food choices to be highly correlated and comparable ((Cheah et al., 2019, Cheah et al., 2020); Persky et al., 2018; Ung et al., 2018). The other studies examined parents' food choices for their child in VR buffet settings (Bouhlal, McBride, Ward, & Persky, 2015; Dolwick & Persky, 2021; Hagerman, Ferrer, Klein, & Persky, 2020; Marcum, Goldring, McBride, & Persky, 2018; McBride, Persky, Wagner, Faith, & Ward, 2013; Persky et al., 2015, 2018; Persky et al., 2020; Persky et al., 2019; Persky & Yaremych, 2020; Yaremych, Kistler, Trivedi, & Persky, 2019). Specifically, they investigated the impacts of providing information about obesity risks for their child and the way of framing the messages or information on parents' food choices for their child in a VR buffet. The impacts of parents' sociodemographic, cognitive, and attitudinal factors on their food choices for their child in a VR buffet were also explored. One study also linked parents' food choices for their child in a VR buffet with their translational movement in the virtual environment (Yaremych et al., 2019). They found that receiving risk information increased parents' lifestyle- and genetics-guilt, and choosing fewer unhealthy foods in the VR buffets reduced both types of guilt. Parents with higher guilt levels reported stronger intentions to improve child-feeding and their own eating behavior in the future. However, guilt could only predict stronger feeding intentions when parents had served their child relatively more unhealthy foods in the VR buffets.

4.2.5. Other applications

There were seven articles that did not fit into any of the aforementioned research domains. Among them, four studies investigated the usability and validity of self-developed virtual environments or VR tools, such as a VR cafeteria and a VR food court (Allman-Farinelli et al., 2019; Isgin-Atici et al., 2020; Jayachandran, Chilakamarri, Coelho, & Mueller, 2017; Karkar et al., 2018). Schnack and colleagues compared the impact of different locomotion approaches (i.e., instant teleportation and motion-tracked walking) in a VR supermarket on participants' shopping behavior (Schnack, Wright, & Holdershaw, 2021). They found that participants who used the two different locomotion approaches did not differ in emotional states and investigated shopping outcomes. A study conducted by Xu and colleagues investigated the comparability of consumers' healthiness perceptions of 20 breakfast cereals in a VR environment and in RL (Xu et al., 2021). They indicated that the data collected in the VR and RL environments were highly comparable. Li and Bailenson (2018) studied the impact of adding the smell and touch of donuts to the virtual environment on participants' satiation and eating behavior. This study found that participants ate significantly fewer donuts and reported higher satiation with the presence of the donut's touch or scent than those without exposure to these cues. However, the study did not draw clear conclusions about the impact of exposing participants to both haptic and olfactory cues.

4.2.6. VR systems

Each of the included articles was then examined to determine what type of VR systems were used. Generally, more than half of the VR systems were HMD systems (n = 48). Desktop VR system (n = 23) was exploited less often than HMD systems but more often than video wall systems (n = 7) and CAVE systems (n = 2). A few studies did not specifically describe or explain which VR system they had used for their studies (n = 10). Some articles used more than one VR systems; therefore, the totally number of VR systems (n = 90) was higher than the number of articles included (n = 85). The summarized results are shown in Fig. 5.

In the research domain that investigated sensory evaluation of food in VR, 15 studies employed HMD systems. One study applied a CAVE system (Pennanen et al., 2020), and video wall systems were applied in four studies (Bangcuyo et al., 2015; Hannum et al., 2019; Hathaway & Simons, 2017; Sinesio et al., 2019). The studies conducted by Hathaway and Simons (2017) used both a video wall system and a desktop VR system. In the research domain that investigated virtual food or food–related cue exposures and responses, thirteen studies employed desktop VR systems, eight studies applied HMD systems, and one study did not explain which VR system was used. There were two studies employing both an HMD system and a desktop VR system (Gutierrez-- Maldonado et al., 2016) or an HMD system and a video wall system (Worch et al., 2020). Among the 14 articles included in the research domain that investigated the application of VR buffets, five studies employed HMD systems (Cheah et al., 2020; McBride et al., 2013; Persky et al., 2018; Ung et al., 2018; Yaremych et al., 2019). The other nine studies did not describe in detail which VR systems were used. In the research domain that investigated the applications of VR supermarkets, six studies used desktop VR systems, ten studies used HMD systems, one study used a CAVE system (Bigné et al., 2016), and another one used a video wall system (Liu et al., 2017). The other two studies used both HMD systems and desktop VR systems (Lombart et al., 2020; Schnack et al., 2019). For detailed information about what VR systems were used and which types of HMDs were employed in the HMD systems, please refer to Table 1 in the supplementary material.

4.3. Comparison between consumers' behavior data collected in VR and RL settings

Studies that examined the comparability of consumer behavior data collected in VR and RL settings are summarized in Table 2. There were five studies that compared VR simulated stores with physical stores or picture simulated stores (Lombart et al., 2020; Pizzi et al., 2019; Siegrist et al., 2019; van Herpen et al., 2016; Waterlander et al., 2015). They all found that consumers' behavior in VR simulated stores was highly comparable to consumers' behavior in physical stores. Compared to picture simulated stores, consumers' behavior in VR stores was also closer to the consumers' behavior in RL (van Herpen et al., 2016). Four studies compared a VR buffet to an RL buffet or a fake food buffet (Cheah et al., 2019)(Cheah et al., 2020); Persky et al., 2018; Ung et al., 2018). These studies suggested that consumers' food choices in a VR buffet and an RL buffet, or in a VR buffet and a fake food buffet, were highly correlated in the amount of kilocalories, grams, carbohydrates and protein they contained. They also found that participants showed similar brain activity when making food choices in an RL buffet and a VR buffet (Cheah et al., 2019). The other two studies compared consumers' perception of the healthiness of breakfast cereals or consumers' sensory evaluation of juices and lemon cakes in a VR environment and an RL environment (Ammann, Stucki, & Siegrist, 2020; Xu et al., 2021). They both found that consumers behavior data collected in VR environments and in RL environments were not statistically significantly different, and consumers' behavior was highly comparable in the two environments.

Some differences between the data collected in the VR and RL settings were also found. Studies found that participants spent more time in VR environments compared to in RL environments (Pizzi et al., 2019; Siegrist et al., 2019; Xu et al., 2021). One study also indicated that

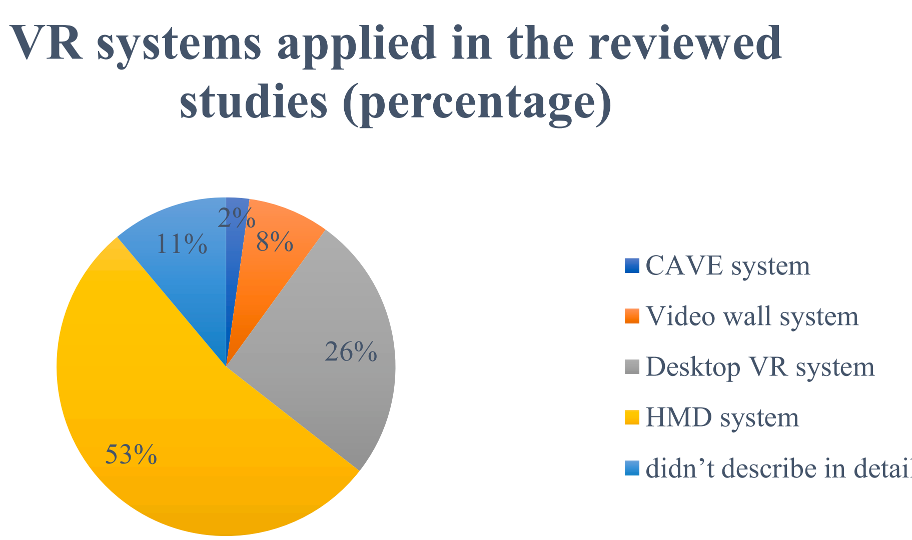

Fig. 5. The virtual reality (VR) systems used in the included studies. HMD: Head-Mounted-Display; CAVE: Cave Automatic Virtual Environment.

Table 2

Summary of the articles that compared consumer behavior data collected in virtual reality (VR) and in real life (RL) settings.

Authors, year	Research domain	Comparison: different experiment settings	Main findings
Ammann, Stucki, and Siegrist (2020)	Sensory evaluation	VR & RL environments	1) It was harder for participants to identify the flavor when the products were shown in modified color (vs. original color); 2) No significant difference between the results obtained in VR and RL environments were found.
Cheah et al. (2019)	VR buffet	VR & RL buffets	1) The total number of calories selected in the VR and RL buffets were highly correlated; 2) When participants chose high- (vs. low-) density food in both VR and RL, the left inferior frontal gyrus showed significant differential activation.
Cheah et al. (2020)	VR buffet	VR & RL buffets	1) Participants' food selections in the VR and RL buffets were significantly and positively correlated in kilocalories, grams, carbohydrates, and protein; 2) The participants perceived the VR buffet as natural, and their selection represented their food selection on an average day.
Lombart et al. (2020)	VR supermarket	VR & RL stores	1) Consumers' perceptions of fruits and vegetables in both fully and low-immersive VR stores are like those in an RL store; 2) Consumers buy more fruits and vegetables in both low and fully immersive VR stores (vs. a RL store); 3) Consumers rely more on extrinsic cues (e.g., price) than intrinsic cues (e.g., appearance) to evaluate food in the fully immersive VR store.
Persky et al. (2018)	VR buffet	VR & RL buffets	Parents' serving amounts and food choices are highly correlated between VR and real buffets.
Pizzi et al. (2019)	VR supermarket	VR & RL stores	1) Consumers reported high levels of perceived realism and presence level in the VR store; 2) Behaviors in the VR-based and physical stores were comparable.
Siegrist et al. (2019)	VR supermarket	VR & RL stores	1) Participants' choice of cereals and the packages they focused on did not significantly differ under VR and RL conditions; 2) Participants checked nutrition information more when had to select a healthy (vs. taste good) cereal in both VR and physical stores; 3) Participants spent more time in front of a VR shelf (vs. a real shelf).
Ung et al. (2018)	VR buffet	VR & fake food buffets	The kJs of the foods chosen in the VR and RL buffets were highly correlated.
van Herpen et al. (2016)	VR supermarket	VR & picture & RL stores	Consumers' behavior in the VR store (vs. picture condition) is more similar to their behavior in the RL store regarding the number of products selected, amount of money spent, and the selection of products from different areas of the shelf.
Waterlander et al. (2015)	VR supermarket	VR & RL stores	1) There was no trend of over-spending in VR; 2) The average presence score was medium-to-high; 3) Shopping patterns in the virtual supermarket were comparable to those in RL.
Xu et al. (2021)	Other	VR & RL environments	1) The perceived healthiness of cereals in VR and RL were highly correlated; 2) The time spent for the task in VR was significantly higher (vs. RL); 3) There was no statistically significant difference between information–seeking behavior in VR and RL; 4) Participants used similar attributes to evaluate the healthiness of the cereals.

consumers bought more food products in VR simulated stores (vs. the RL store), and they relied on extrinsic cues (i.e., prices) in the VR simulated stores rather than on the intrinsic cues (i.e., appearance) of food products they used in the physical store (Lombart et al., 2020). However, this result was only tested with fruits and vegetables.

5. Discussion

The present study systematically reviewed the existing literature on the application of VR technology in food and consumer behavior research with the aim of understanding and exploring the validity of VR and the limitations and possibilities of applying VR technology in this research field. The literature search and screening resulted in 85 articles that were included in the final data extraction, syntheses, and analyses. Subsequently, the three main research objectives were achieved. To facilitate reading, the following discussion is divided into subsections according to the research domains. A subsection that proposes gaps and future studies is presented last.

5.1. VR and food sensory evaluation research

Based on the results of this systematic review, the application of VR in food sensory evaluation mainly focused on: 1) exploiting VR to provide contextual information or environments for food sensory evaluation; and 2) studying the impact of the simulation of the color of food on perception and evaluation. One advantage of using VR in food sensory evaluation research is that highly realistic contextual information can be designed and delivered in fully controlled and highly standardized laboratory settings. In VR contextual environments, participants feel like they are drinking a beer or coffee in a bar or a coffee house, for example, rather than doing a scientific experiment. Meanwhile, environmental factors that might influence the study results in RL can be controlled. Some disadvantages of traditional sensory booths (e.g., that the tasting environment is different from a real consumption context) can be reduced.

The studies also indicated that the color of food or food packages can be successfully simulated in VR to affect the perception and evaluation of food products and brands (Ammann, Stucki, & Siegrist, 2020; Huang et al., 2019; Huang et al., 2021; Huang & Wan, 2019; Sugita et al., 2017; Wang et al., 2020). This implication is particularly relevant for the development and marketing of new food products and brands. Virtual prototypes can be pre-examined and adjusted before real products are produced, saving time and costs. This result also offers significant potential for innovative experimental designs in food consumer behavior research. The application of VR broadens the range of possible research topics, from only investigating products or scenarios that already exist in RL to testing those that may not exist in RL but can be specifically designed for a research question. However, the reviewed studies only investigated one sensory characteristic (i.e., color). Future studies can focus on exploring other visual characteristics, such as shape and size. For the simulation of other sensory characteristics of food, such as taste and texture, VR might not be the appropriate tool to use. Other technologies, such as augmenting flavor with electric sensors, can be used to simulate the taste of the food (Ranasinghe, Jain, Karwita, & Do, 2017). It would be interesting for future studies to explore the potential of combining VR and those technologies and applying them to sensory evaluation research.

Highly immersive VR systems were mainly used in this research domain. Even though the HMD system was the most frequently employed, this might not be the best VR system for sensory evaluation studies. Because of the characteristics of HMD systems, participants were blinded to the food samples presented in RL. This situation could cause difficulties in tasting and evaluating the samples (Huang et al., 2019). The video wall system may be a better choice for sensory evaluation studies because participants can be immersed in contextual environments without being totally blinded to the food samples presented in the real world. However, HMD systems can be beneficial if participants can interact with the sample without actually seeing it. For example, trackers could be used to map the position of the real food sample in RL to the position of its 3D model in VR. In this way, participants could interact with the real food sample by interacting with the 3D model (Nivedhan et al., 2020; Wang et al., 2020).

5.2. VR and food consumer behavior research

Of the 85 reviewed articles, 34 investigated the applications of VR to the observation of consumers' shopping behavior and food choices in environments with contextual information of point-of-sale (i.e., store or supermarket) and point-of-consumption (i.e., buffet). As innovative alternatives to physical supermarkets and RL buffets, VR supermarkets and VR buffets are preferred, especially when examining those variables that cannot be easily manipulated or controlled in traditional test environments, such as food shape, packaging, colors, and labels, as well as environmental factors (Bigné et al., 2016; Blitstein et al., 2020; Goedegebure et al., 2020; Liu et al., 2017; Lombart et al., 2019, Lombart et al., 2020; Verhulst et al., 2017, Verhulst et al., 2018; Zhao et al., 2017). Another advantage of using VR in food consumer behavior studies is that methods such as nudging and pricing can be more time- and cost-efficiently implemented in VR supermarkets and VR buffets than in RL settings (Blom et al., 2021; Hoenink et al., 2020). Meanwhile, virtual environments are computer generated; thus, the same experimental environments can be ensured as long as the same VR system and hardware are provided. Therefore, the same experiment can be conducted in different places with participants from varied cultural or ethnic backgrounds. This might be advantageous, for example, for multinational companies. Food development and marketing strategies can be formulated according to the specific preferences of consumers in specific markets.

In addition, VR can be combined with other technologies, such as eye-tracking, body movement tracking, and neurophysiological response measurement (Bigné et al., 2016; Cheah et al., 2019; Huang et al., 2021; Siegrist et al., 2019; Yaremych et al., 2019). Together with other technologies, VR can help researchers gain a deeper knowledge of the mechanisms behind consumers' behavior and ultimately help consumers make better and healthier food choices. VR provides an environment that can easily implement technologies such as eye-tracking. Compared to eye-tracking in RL, VR provides the opportunity to observe and calculate participants' gaze behavior in a 3D space during the session; attentional data such as the area of interest can be easier defined and tracked in VR as well (Clay, Koenig, & Koenig, 2019). The convenience of implementing these technologies in VR also broadens the potential applications of VR in food consumer behavior research.

However, using VR for food consumer behavior research also has some limitations. First, virtual environments may lack some of the characteristics of real food choices or consumption environments, such as the touch, weight, or smell of foods and the presence of other people (Allman-Farinelli et al., 2019; Kuo et al., 2016; Lombart et al., 2019; Marcum et al., 2018; Persky et al., 2018). Therefore, the influence of these factors on consumers' food choices and consumption might not be considered or investigated in most studies. Therefore, VR might not be the best tool to investigate consumer's actual patterns of food consumption in RL. Second, the nature of VR implies that compared to the real world, participants may need more time to get familiar with the virtual environment before starting the experiment so that they will focus on the task itself instead of on exploring the novel and unfamiliar environments (Pizzi et al., 2019; Siegrist et al., 2019; Xu et al., 2021). Especially with VR systems that have low immersion levels, the experience might be less natural and further away from an RL experience.

Nevertheless, the validity of VR tools for food consumer behavior research was supported by the result that consumer behavior data collected in VR settings (e.g., VR supermarkets or VR buffets) were highly correlated and comparable to the data collected in RL settings (e.g., physical stores or RL buffets) (Ammann, Stucki, & Siegrist, 2020; Cheah et al., 2020, 2019; Lombart et al., 2020; Persky et al., 2018; Pizzi et al., 2019; Schnack et al., 2019; Siegrist et al., 2019; Ung et al., 2018; van Herpen et al., 2016; Waterlander et al., 2015; Xu et al., 2021). The validity was further supported by the evidence showing that similar brain activities of participants were found when they were making food choices in a RL and a VR buffet (Cheah et al., 2019). Therefore, despite the potential limitations of VR, it is clear that VR has significant potential to provide effective and innovative methods and study designs for food consumer behavior research.

5.3. VR and food cue exposure and responses

Virtual food or food-related cues have been used to study food–cue exposure and response in the general population and in patients with eating disorders. The reviewed studies demonstrated that virtual food or food-related cues can successfully induce eating-behavior-related emotional reactions in consumers (Ammann, Hartmann, et al., 2020; Ferrer-Garcia, Gutierrez-Maldonado, & Pla, 2013; Ferrer-García, Gutiérrez-Maldonado et al., 2017; Ferrer-Garcia, Gutierrez-Maldonado, Treasure, & Vilalta-Abella, 2015; Gorini et al., 2010; Gutierrez-Maldonado et al., 2016; Gutiérrez-Maldonado et al., 2006; Kuo et al., 2016; Ledoux et al., 2013; Pla-Sanjuanelo et al., 2019; Tuanquin et al., 2018). However, studies yielded inconsistent results about the extent to which VR cues are more effective than picture cues, which were commonly applied or recognized in studies that investigated food cue exposures and responses (Gorini et al., 2010; Ledoux et al., 2013; van der Waal et al., 2021). The possibility that picture cues are sufficiently effective may raise the question of why we need more expensive VR systems when pictures are as effective. Even though the effectiveness of virtual cues (vs. pictures) is still up for discussion, VR may be preferable to pictures cues because it has greater potential to provide food-related contextual information. At the same time, compared to real food cues, VR has the advantage of ensuring that the participants will be exposed to exactly the same food cues even at different times and locations. However, additional studies need to be conducted to explore whether VR cues are better than pictures and real food cues to provide suggestions for which cues to use in food–cue exposure and response studies.

5.4. Existing gaps in the literature and future studies

This paper systematically reviewed the articles that employed VR in food and consumer behavior research. The results indicate some gaps or limitations in the literature. First, most of the reviewed studies were conducted in the US and Europe. Children, males, and older populations were under represented in the reviewed studies. Therefore, additional studies can be conducted in more varied cultural and ethnic environments with more focus on these populations. Second, limited food categories were tested in the reviewed studies. Future studies should aim at exploring a broader set of food categories, especially with the food products that are unpackaged and whose consumption is less context-dependent, such as fruit, vegetables, bakery bread, dried fruits, nuts, and fresh meat and meat products. Third, only a limited number of studies have contributed to the investigation of the validity of VR. Although the results of these studies have justified the validity of VR as a tool for collecting consumer behavior data, additional studies that aim at investigating the comparability of the behavior data collected in VR and RL settings will further support the conclusion.

Furthermore, most of the VR systems or applications applied in the reviewed studies only provided one sensory modality (i.e., visual) to consumers; some sensory modalities (e.g., olfactory, haptic, and auditory modalities) from a food consumption or selection environment in RL were missing. Future studies could investigate whether adding these sensory modalities to VR environments would influence consumer behavior. In particular, studies could focus on understanding what level of sensory modalities would be sufficient for consumers to feel immersed and present in the VR environment and behave similarly to the way they behave in RL food consumption or selection environments. Studies could also investigate the impact of different ways of interaction and locomotion in VR systems on the results of food and consumer behavior studies. Meanwhile, more studies are needed to help understand the impact of the immersion level of VR systems on consumer behavior because inconsistent results on this topic have been found in the literature (Lombart et al., 2020; Peukert et al., 2019; Schnack et al., 2019).

Finally, studies in sensory evaluation have provided support for the ability of applying VR to simulate the color of foods (Ammann, Stucki, & Siegrist, 2020; Huang et al., 2019; Huang et al., 2021; Huang & Wan, 2019; Sugita et al., 2017; Wang et al., 2020). Given the significant results from food sensory evaluation studies, it would also be interesting to investigate the impacts of the simulation of foods' visual characteristics (e.g., color, shape, and size) on food choice and shopping behavior. Future studies should also explore and use the full potential of combining a VR system with eye-tracking and other systems to investigate the link between neurophysiological or attentional activities and consumer behavior.

6. Conclusion

The present systematic review summarizes the recent scientific evidence regarding the applications of VR in food and consumer behavior research to explore the validity of VR and the possibilities and limitations of using VR technology in this research field. The results indicate that VR has been widely used in food and consumer behavior research. It has been employed in research domains that investigated sensory evaluation of food in VR contextual environments, food choice in VR buffets, shopping behavior and product perception in virtual simulated stores, and emotional and physical responses to virtual food or food–related cues or environments. Evidence suggests that VR is a valid tool for research that investigates consumer behavior towards food. VR supermarkets and VR buffets can be used as innovative alternatives to RL tools. VR food cues can also successfully induce food–related emotional responses. However, more evidence might be needed to better understand whether VR food cues can always provide better results than picture cues. On the basis of the results of this paper, the authors recommend that future studies should be conducted with a broader set of food categories. Additional studies that investigate the comparability of consumers' behavior in VR and RL would be welcome. Future studies can also focus on investigating the impact of providing more sensory modalities than merely visuals on the results of food and consumer behavior studies.

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Declaration of interest

None.

Author statement

Chengyan Xu: methodology, investigation, data curation, formal analysis, writing-original draft. Michael Siegrist: methodology, writing-review & editing, validation, supervision. Christina Hartmann: conceptualization, methodology, writing-review & editing, validation, supervision.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tifs.2021.07.015.

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